

Table 4 — DeepStack search-space ablation

Step	Technique/search-space addition	BS=4 STPS	BS=1024 STPS
1	AstraSim (tp/pp/dp)	177.1	5,729.2
2	+ Full Parallel	256.4 (+45%)	21,252.3 (+271%)
3	+ Flexible Parallelism Across Module	256.4 (+0%)	24,487.8 (+15%)
4	+ Search On-chip Architecture	314.2 (+23%)	31,349.9 (+28%)
5	+ Comm-Comp Overlap	340.5 (+8%)	38,061.0 (+21%)
6	+ Stacking DSE	493.3 (+45%)	51,094.5 (+34%)
7	+ NoC DSE	494.1 (+0%)	54,280.3 (+6%)
	Ablation gain	2.79×	9.47×